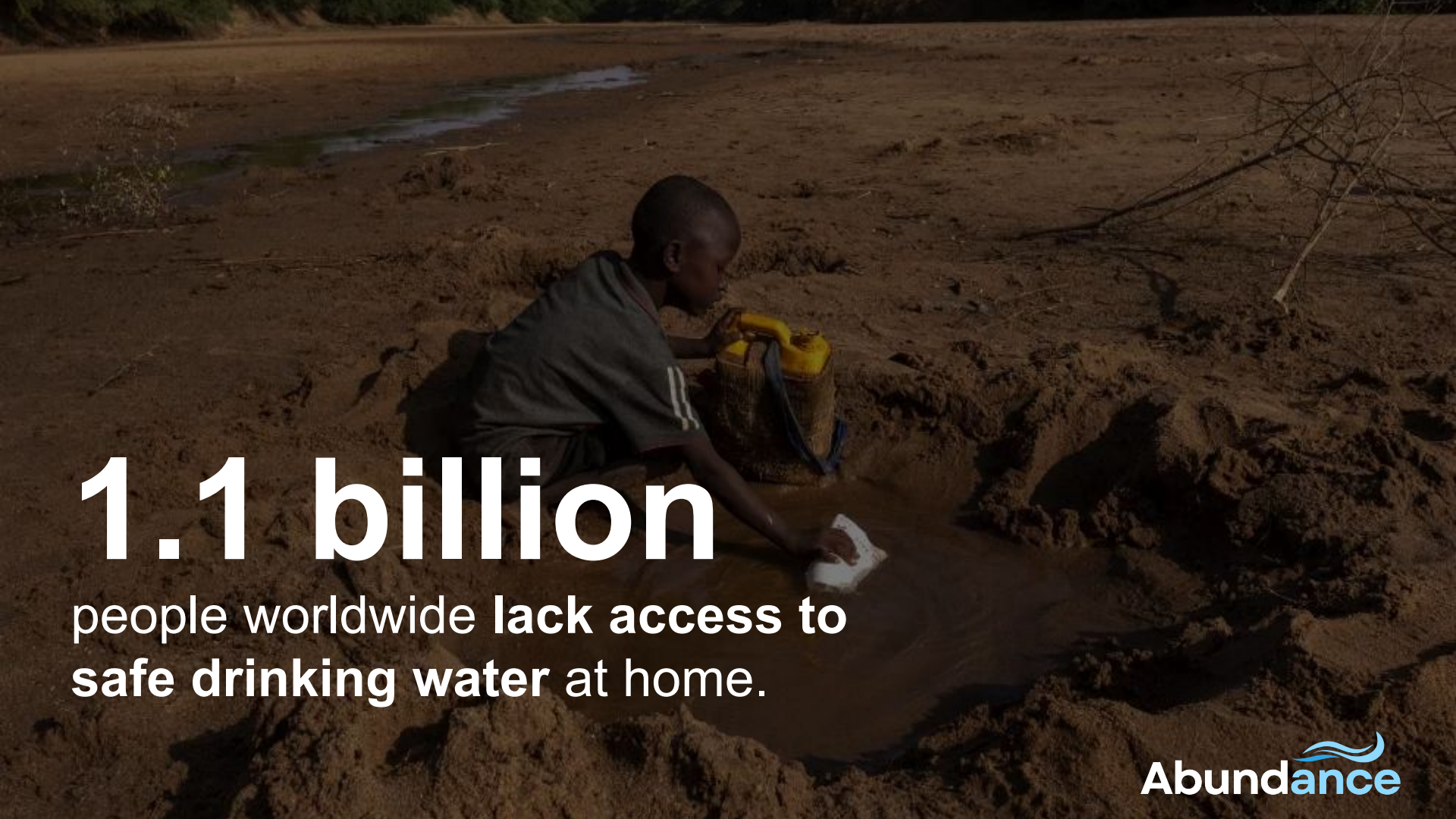

Abundance
From water scarcity to abundance: Revolutionizing access to clean water worldwide, one community at a time


1.1 billion
people worldwide **lack access to safe drinking water** at home.
Abundance


In Guinea region, **42.8%** of the population lack clean drinking water at home.


Air
Water
Abundance


77.9%
average humidity
year-round
Abundance



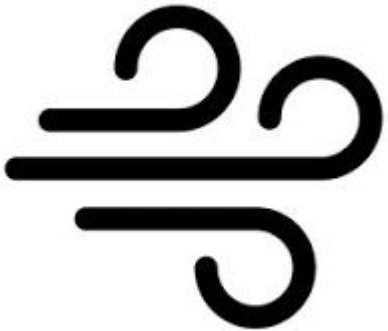
AWG

Air Water Generation

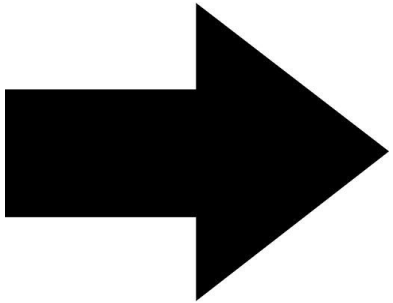
Abundance

AWG process

1



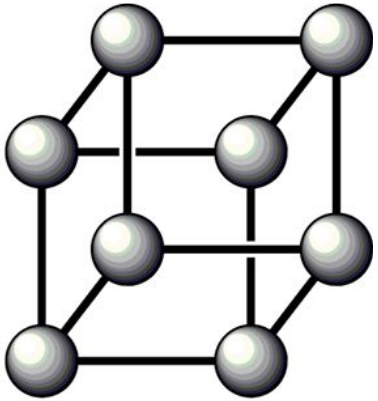
2



3



Existing desiccants



MOFs



Silica Gels



Zeolites



Issues with current desiccants

1

Challenges in quantity extracted

2

Require significant amounts of energy

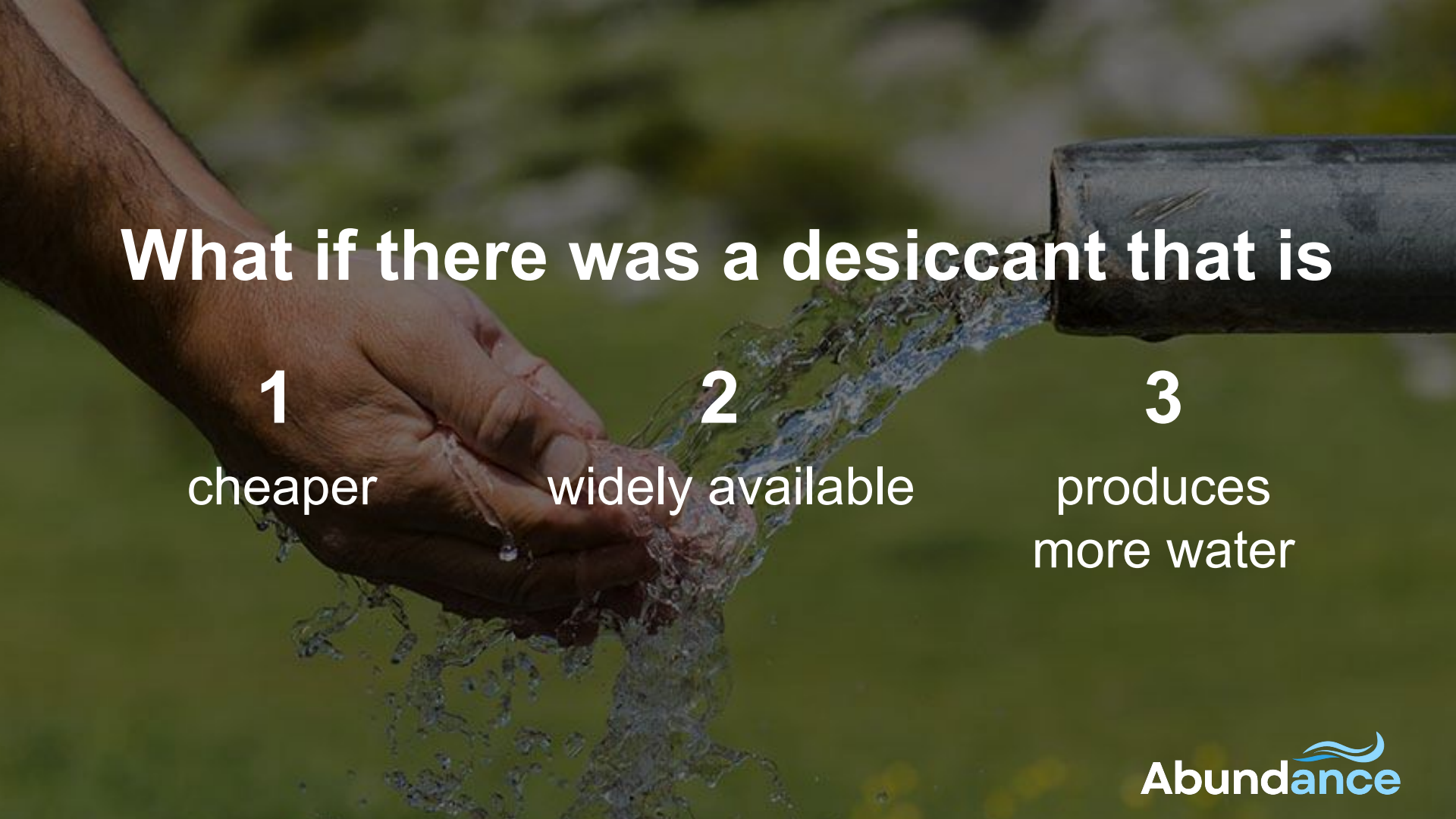
3

Desiccant itself is inaccessible

Abundance

Cost per litre
needs to be lowered





What if there was a desiccant that is

1
cheaper

2
widely available

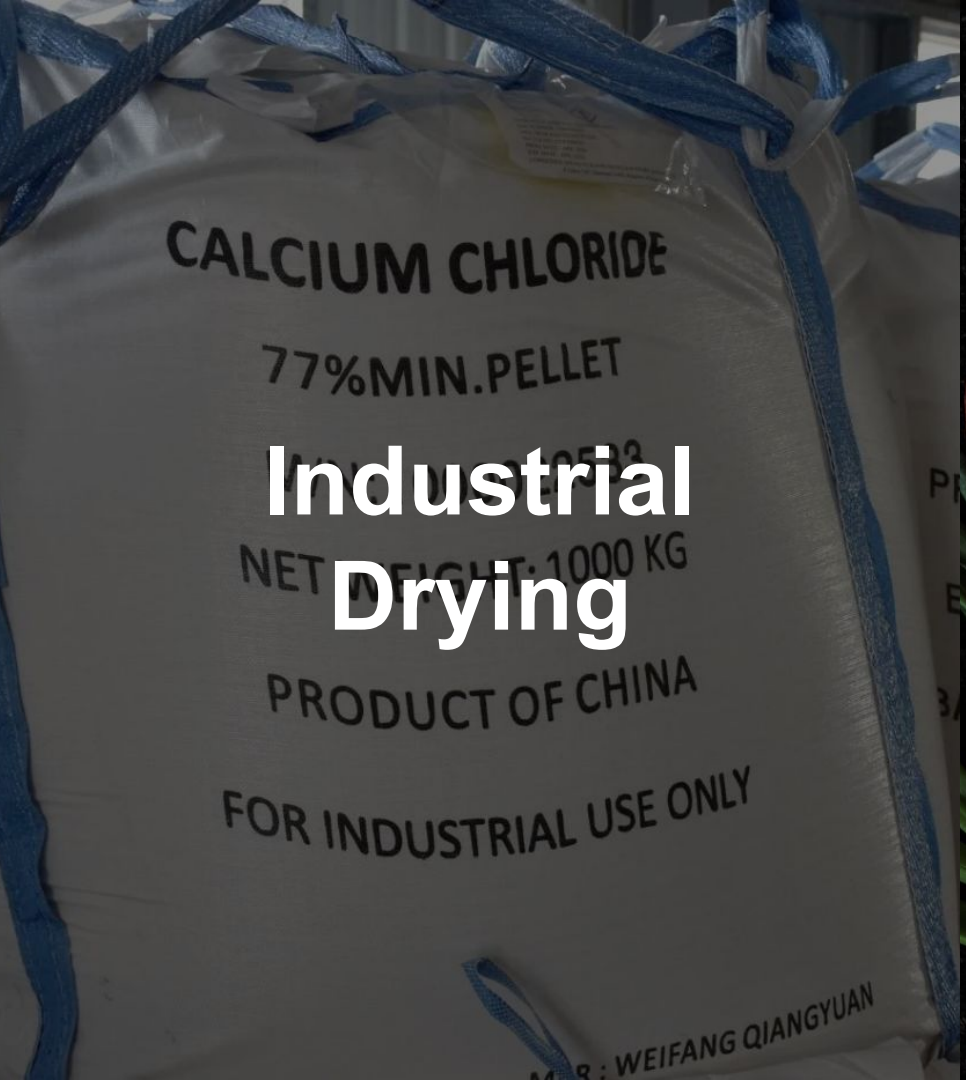
3
produces more water

Abundance

Calcium chloride injected Cellulose sponges



Abundance



Industrial Drying

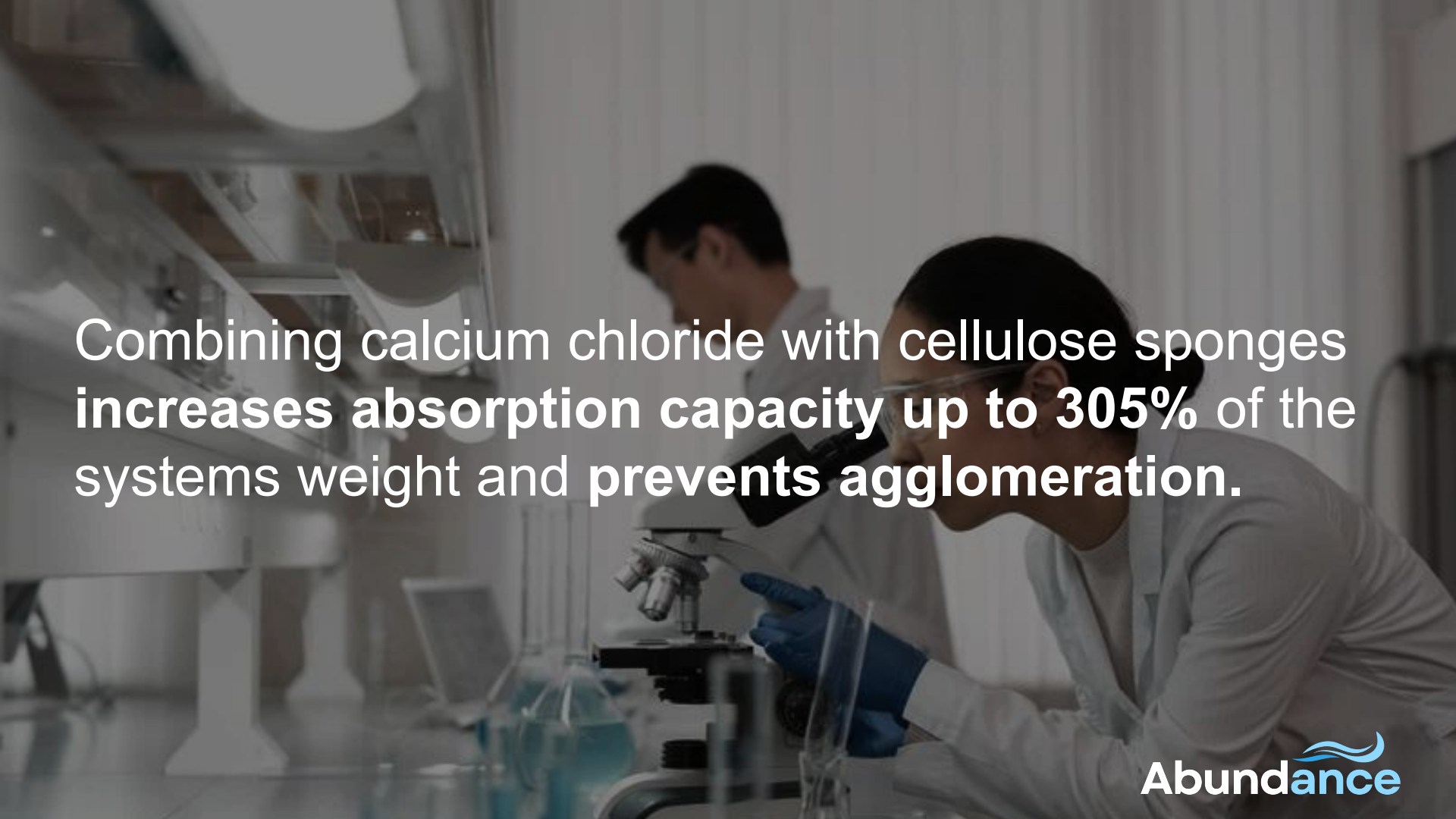


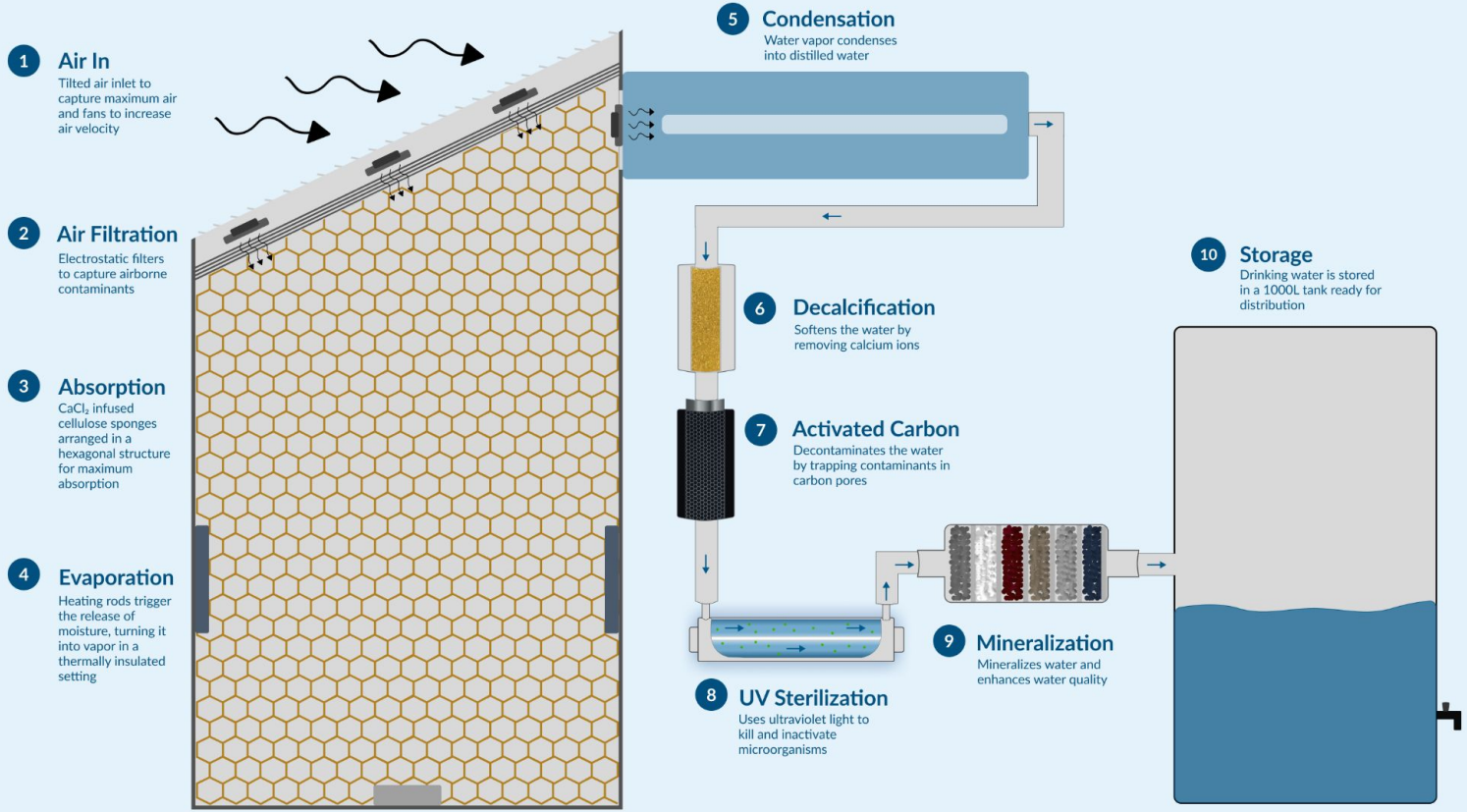
Food Packaging

Abundance

Why Cellulose Sponges?

Abundance


Combining calcium chloride with cellulose sponges **increases absorption capacity up to 305%** of the systems weight and **prevents agglomeration.**
Abundance


1 Air In
Tilted air inlet to capture maximum air and fans to increase air velocity
2 Air Filtration
Electrostatic filters to capture airborne contaminants
3 Absorption
$CaCl_2$ infused cellulose sponges arranged in a hexagonal structure for maximum absorption
4 Evaporation
Heating rods trigger the release of moisture, turning it into vapor in a thermally insulated setting
5 Condensation
Water vapor condenses into distilled water
6 Decalcification
Softens the water by removing calcium ions
7 Activated Carbon
Decontaminates the water by trapping contaminants in carbon pores
8 UV Sterilization
Uses ultraviolet light to kill and inactivate microorganisms
9 Mineralization
Mineralizes water and enhances water quality
10 Storage
Drinking water is stored in a 1000L tank ready for distribution

1

Air In

Tilted air inlet to capture maximum air and fans to increase air velocity

2

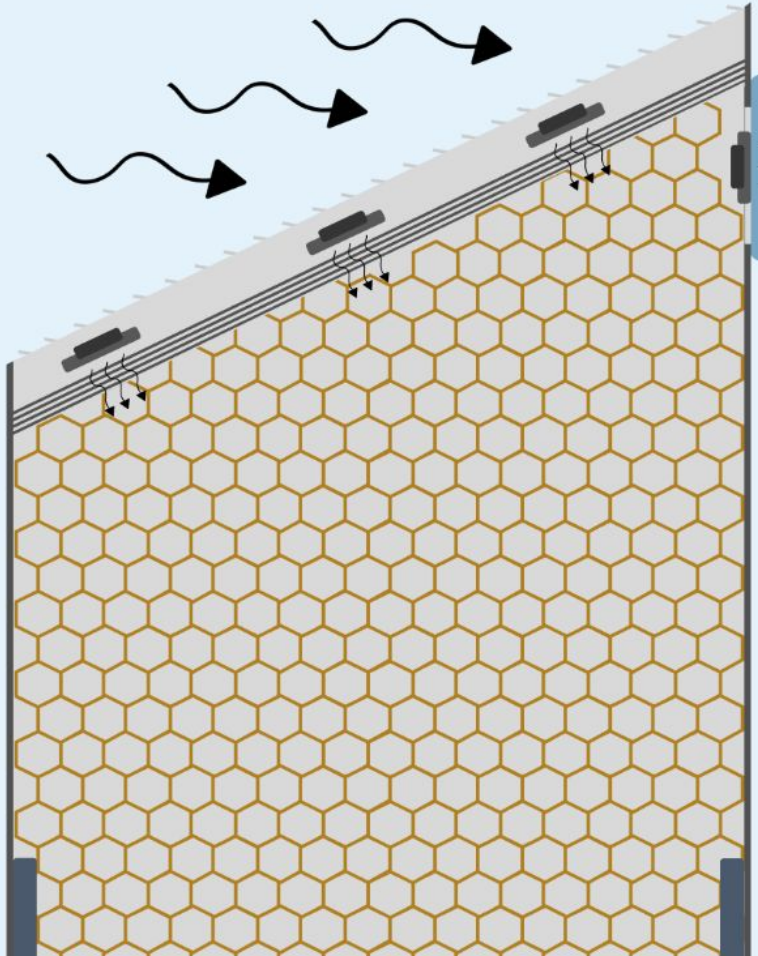
Air Filtration

Electrostatic filters to capture airborne contaminants

3

Absorption

$CaCl_2$ infused cellulose sponges arranged in a hexagonal structure for maximum absorption



4

Evaporation

2

Air Filtration

Electrostatic filters to capture airborne contaminants

3

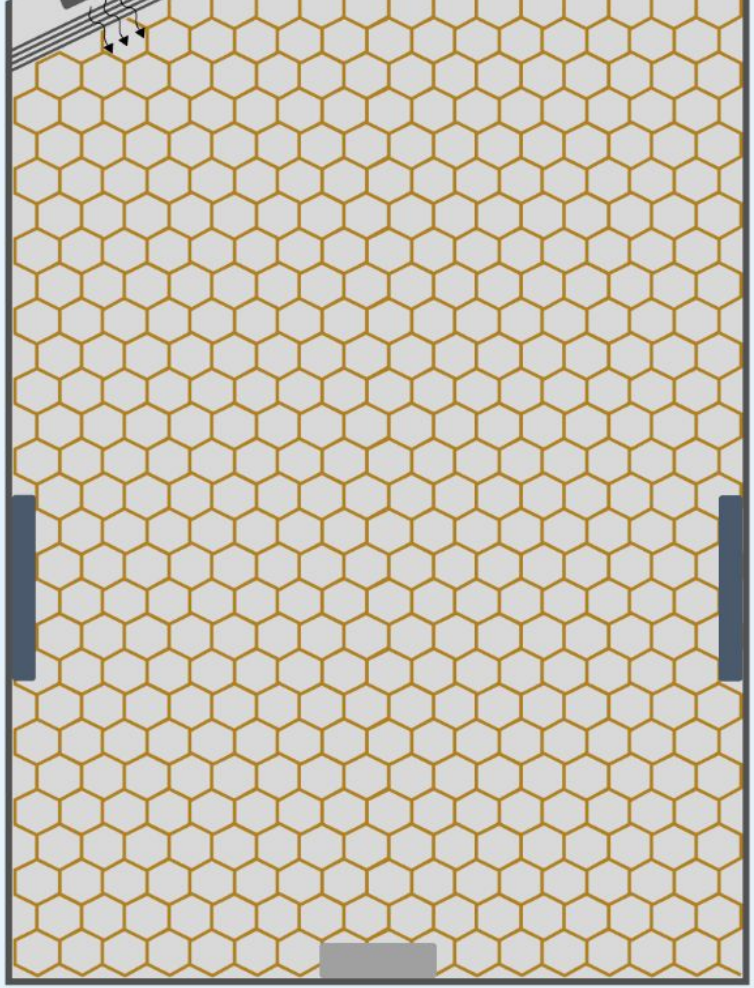
Absorption

$CaCl_2$ infused cellulose sponges arranged in a hexagonal structure for maximum absorption

4

Evaporation

Heating rods trigger the release of moisture, turning it into vapor in a thermally insulated setting





5 Condensation

Water vapor condenses into distilled water

6 Decalcification

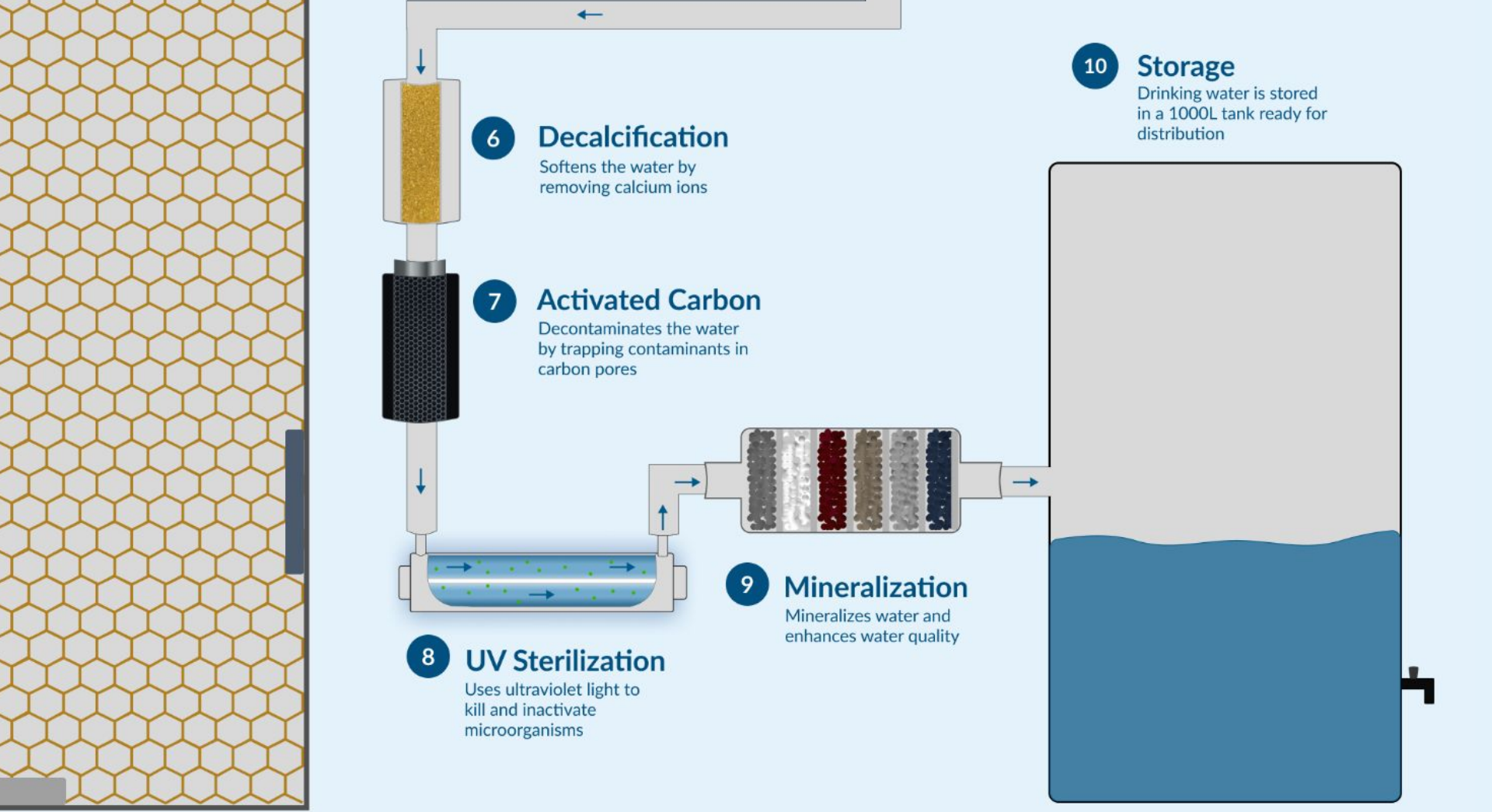
Softens the water by removing calcium ions

7 Activated Carbon

Decontaminates the water by trapping contaminants in carbon pores

10 Storage

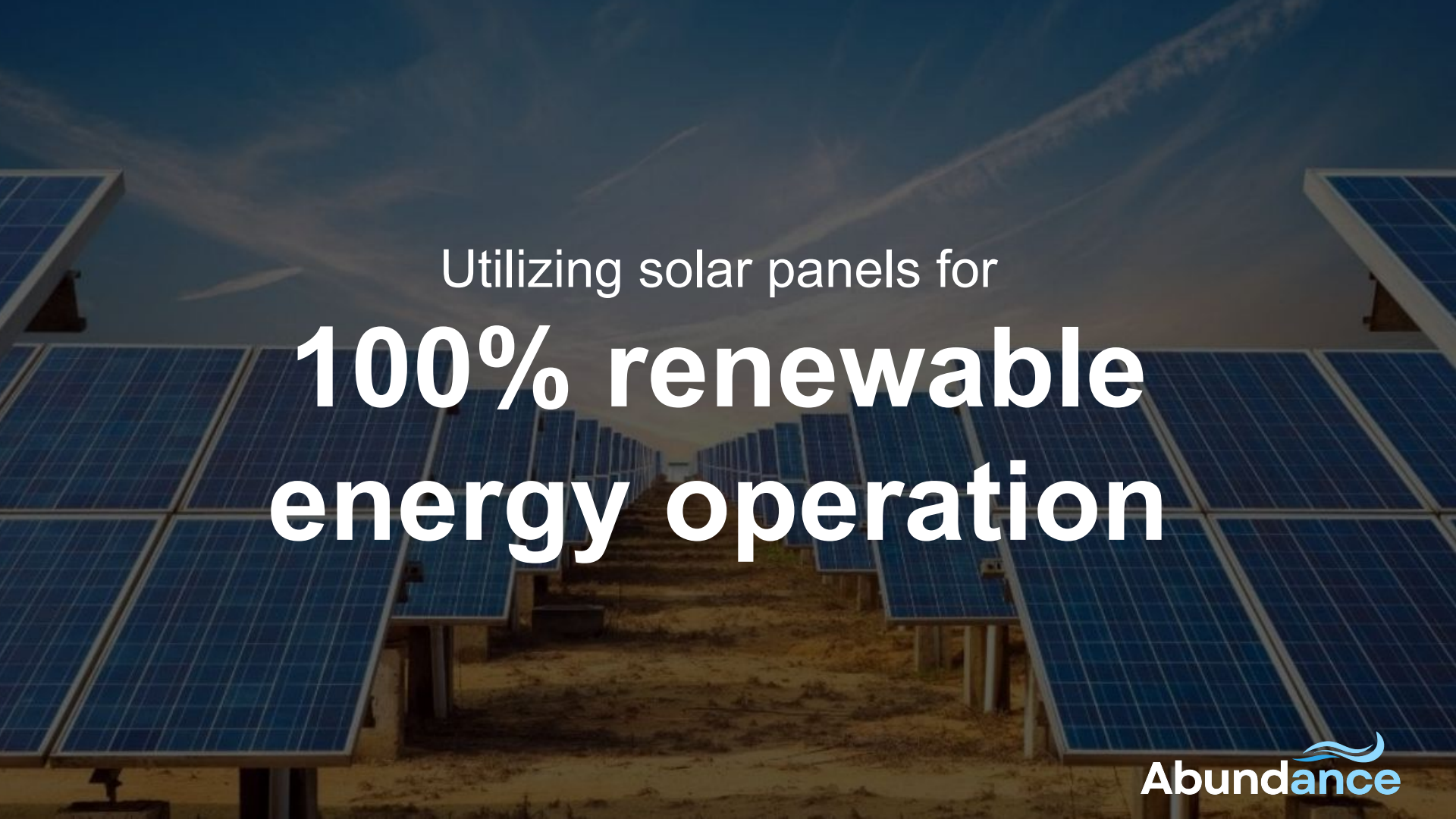
Drinking water is st
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distribution


10
Storage
Drinking water is stored in a 1000L tank ready for distribution
6
Decalcification
Softens the water by removing calcium ions
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Activated Carbon
Decontaminates the water by trapping contaminants in carbon pores
9
Mineralization
Mineralizes water and enhances water quality
8
UV Sterilization
Uses ultraviolet light to kill and inactivate microorganisms


Our system produces
1000 liters
of clean drinking water daily,
benefiting 200-250 people.
Abundance


Initial costs high but projected to decrease to
2 cents per liter
within one year.
Abundance


7-8x cheaper
than industry standard
Abundance


Utilizing solar panels for
100% renewable energy operation
Abundance


92 times
more energy efficient
than industry standard
Abundance

We are not the only ones who think this is a great idea….



Joakim Carlson
COO at Absorbtech



Jackson Lord
Past Program Director at X moonshot factory of AWG technology



Bill Meeker
Board member at El Hogar Projects



Jackson Lord
Past Program Director at X moonshot factory of AWG technology

> ***A low-cost desiccant like calcium chloride*** *is especially a great solution for communities that need a centralized solution... This device could even be* ***scalable*** *to South Asia and Latin America.*

Abundance shaping a future where clean water is a **fundamental human right.**



L’chelle Bester



Jhanvi Matta



Youssef Kusibati



Maryam Imran